

NAGAPATTINAM, India

WMO: 433470

Lat: 10.767N

Lon: 79.850E

Elev: 9

StdP: 101.22

Time Zone: 5.50 (IND)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 22.0	(c) 22.8	(d) 17.4	(e) 12.4	(f) 27.0	(g) 18.4	(h) 13.3	(i) 27.0	(j) 5.5	(k) 26.5	(l) 5.2	(m) 26.6	(n) 1.9	(o) 320	(p) 0.372

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	8.5	37.7	26.2	36.9	26.2	36.0	26.2	29.5	33.4	29.0	32.9	28.7	32.4	3.4	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
28.5	24.9	31.6	28.1	24.3	31.4	27.7	23.7	31.1	97.2	34.0	95.1	33.0	93.3	32.4	31.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.0	5.3	4.7	20.6	40.6	1.0	1.8	19.9	41.9	19.3	42.9	18.7	43.9	17.9	45.2		
			18.8	30.4	1.0	0.6	18.1	30.9	17.5	31.2	16.9	31.6	16.2	32.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	29.4	26.2	27.1	29.1	31.2	31.9	32.1	31.6	30.6	30.3	28.9	27.1	26.3
	DBStd	2.53	1.10	1.16	1.14	1.11	1.66	1.50	1.22	1.60	1.39	1.66	1.31	1.23
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	7072	501	479	593	636	678	664	669	638	608	586	514	505
	CDD18.3	4031	243	246	334	386	420	414	411	380	358	328	264	247
	CDH23.3	48952	2042	2323	3972	5398	5894	5769	5578	4881	4511	3841	2597	2146
(14) Wind	WSAvg	2.1	2.0	1.7	1.5	1.9	2.5	2.9	2.6	2.3	2.1	1.4	1.8	2.1
	PrecAvg	1243	37	19	21	29	48	38	41	78	85	218	410	228
	PrecMax	1770	199	91	186	226	188	154	126	242	203	737	844	458
	PrecMin	624	0	0	0	0	0	0	11	9	1	0	18	41
	PrecStd	290	42	28	44	49	49	46	30	54	49	153	207	157
	DB	30.2	32.0	34.5	37.5	38.9	38.9	38.0	37.6	36.9	35.0	32.2	30.4	27.2
	MCWB	25.0	26.1	26.6	27.2	26.2	26.7	26.0	25.5	25.0	26.1	26.2	25.5	25.5
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	29.5	30.9	33.1	35.4	37.4	37.7	36.9	36.3	35.4	33.6	31.0	29.6	29.6
	MCWB	24.6	25.7	26.8	27.8	26.9	26.2	25.8	25.5	25.5	26.5	26.2	25.4	25.4
	DB	28.9	30.2	32.5	34.4	36.2	36.7	36.0	35.2	34.2	32.6	30.1	29.1	29.1
	MCWB	24.2	25.0	26.6	28.0	27.3	26.2	25.8	25.8	26.0	26.7	26.1	25.1	25.1
	DB	28.3	29.5	31.7	33.7	34.9	35.7	34.9	34.0	33.2	31.7	29.3	28.5	28.5
	MCWB	23.9	24.6	26.3	27.8	28.0	26.2	26.1	26.1	26.5	26.7	25.9	24.9	24.9
	WB	26.6	27.1	28.5	29.6	30.4	29.8	29.2	29.4	28.7	28.5	27.7	27.2	27.2
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCDB	29.1	30.5	32.3	34.3	33.9	34.8	33.0	32.8	32.3	32.0	30.1	28.8	28.8
	WB	25.8	26.4	27.7	29.1	29.6	29.0	28.5	28.6	28.1	27.8	27.2	26.6	26.6
	MCDB	28.1	29.6	31.3	33.5	33.2	32.8	32.1	32.1	31.7	31.1	29.4	28.4	28.4
	WB	25.3	26.0	27.3	28.6	29.1	28.4	28.0	27.9	27.6	27.5	26.8	26.2	26.2
	MCDB	27.6	29.1	30.9	32.8	32.7	32.0	31.4	31.4	31.1	30.6	28.9	27.9	27.9
	WB	24.8	25.5	26.9	28.2	28.7	27.9	27.5	27.4	27.3	27.2	26.5	25.7	25.7
	MCDB	27.2	28.5	30.5	32.2	32.3	31.5	31.2	30.8	30.6	30.1	28.5	27.5	27.5
(35) Mean Daily Temperature Range	MDBR	5.0	5.6	5.8	5.9	7.1	8.5	8.1	7.8	7.2	5.5	4.2	4.0	4.0
	5% DB	5.3	5.7	6.5	6.6	8.6	9.4	9.1	9.3	8.4	6.9	5.3	4.7	4.7
	MCWBR	2.5	2.4	2.3	2.2	2.8	3.1	3.0	2.9	2.7	2.4	2.2	2.3	2.3
	5% WB	4.6	5.1	5.8	6.2	7.0	8.7	8.2	7.6	7.1	6.0	4.5	4.1	4.1
	MCWBR	2.6	2.4	2.4	2.5	2.8	3.5	3.4	3.4	2.9	2.5	2.4	2.5	2.5
(40) Clear-Sky Solar Irradiance	taub	0.455	0.446	0.481	0.496	0.581	0.517	0.525	0.514	0.502	0.519	0.503	0.494	0.494
	taud	2.257	2.271	2.178	2.193	1.955	2.161	2.131	2.169	2.205	2.151	2.172	2.161	2.161
	Ebn at Noon	851	873	846	825	743	782	779	799	816	798	800	803	803
	Edh at Noon	136	139	155	151	188	151	156	153	148	154	147	147	147
(44) All-Sky Solar Radiation	RadAvg	5.02	5.91	6.40	6.32	6.01	5.75	5.61	5.72	5.69	4.84	4.12	4.21	4.21
	RadStd	0.28	0.26	0.30	0.30	0.37	0.33	0.29	0.28	0.34	0.31	0.43	0.35	0.35

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	+0.28	N/A	N/A	+0.86	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page